Exponentially Convergent Modular Adaptive Control for Nonlinear Target Tracking

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***Abstract*—A**

***Index Terms*—Neural networks, Adaptive control, Stability of nonlinear systems**

I. Introduction

Depth with nonlinearity creates no bad local minima in ResNets [1]. CHANGE ALL C2 to LIPSCHITZ. YOUR EPSILON ARGUMENT ARE WRONG.

II. Preliminaries

A. Notation

Let $n \in \mathbb{Z}_{\geq 1}$. The $\|\|$ means 2-norm. A continuous function $\mathcal{C}^2$...Zero vector. $\subsetneq$. Ball or radius r. Closure. Filippov regularization. Signum. C^1 norm.

B. Nonsmooth Analysis

See latest Cristian paper.

C. Deep Residual Neural Network Model

The residual neural network (ResNet) architecture, characterized by skip connections and hierarchical feature extraction, models incremental changes rather than complete transformations of the underlying nonlinear mapping. This architecture learns the differences (or "residuals") between the input and desired output at each layer, thereby enabling effective function approximation for complex nonlinear systems without requiring explicit governing equations. A fully-connected feedforward ResNet is constructed as follows.

Let $b \in \mathbb{Z}_{\geq 0}$ be the number of building blocks, let the input be $x \in \mathbb{R}^{L_{\text{in}}}$, and let the output be $y \in \mathbb{R}^{L_{\text{out}}}$, where $L_{\text{in}}, L_{\text{out}} \in \mathbb{Z}_{\geq 1}$. For each block index $i \in \{0, \ldots, b\}$, let $k_i \in \mathbb{Z}_{\geq 1}$ denote the number of hidden layers in the i^{th} block, let $\kappa_i \in \mathbb{R}^{L_{i,0}}$ denote the block input, and let $\theta_i \in \mathbb{R}^{p_i}$ denote the vector of parameters (weights and biases) associated with the i^{th} block. Note that $\kappa_0 \triangleq x$ and $L_{0,0} \triangleq L_{\text{in}}$.

For all $(i,j) \in \{0, \ldots, b\} \times \{0, \ldots, k_i + 1\}$, let $L_{i,j} \in \mathbb{Z}_{\geq 1}$ denote the number of neurons in the j^{th} layer for the i^{th} block. For all $(i,j) \in \{0, \ldots, b\} \times \{0, \ldots, k_i\}$, define the augmented dimension $L_{i,j}^a \triangleq L_{i,j} + 1$. Each block function $\Phi_i : \mathbb{R}^{L_{i,0}^a} \times \mathbb{R}^{p_i} \to \mathbb{R}^{L_{i,k_i+1}}$ is a fully connected, feedforward deep neural network (DNN) with $L_{i,k_i+1} \triangleq L_{\text{out}}$ for all $i \in \{0, \ldots, b\}$. For any input $v \in \mathbb{R}^{L_{i,j}^a}$, the DNN is defined recursively by

$$\varphi_{i,j}(v) \triangleq \begin{cases} V_{i,0}^{\top} v, & j = 0, \\ V_{i,j}^{\top} \phi_{i,j}(\varphi_{i,j-1}(v)), & j \in \{1, \ldots, k_i\}, \end{cases} \tag{1}$$

with $\Phi_i(v, \theta_i) = \varphi_{i,k_i}(v)$.

For each $j \in \{0, 1, \ldots, k_i\}$, the matrix $V_{i,j} \in \mathbb{R}^{L_{i,j}^a \times L_{i,j+1}}$ contains the weights and biases; in particular, if a layer has n inputs (including the appended bias) and the subsequent layer has m nodes, then $V \in \mathbb{R}^{n \times m}$ is constructed so that its $(r,c)^{\text{th}}$ entry represents the weight from the r^{th} node of the input to the c^{th} node of the output, with the last row corresponding to the bias terms. For the DNN architecture described by (1), the vector of DNN weights of the i^{th} block is $\theta_i \triangleq \begin{bmatrix} \text{vec}(V_{i,0})^{\top} & \cdots & \text{vec}(V_{i,k_i})^{\top} \end{bmatrix}^{\top} \in \mathbb{R}^{p_i}$, where $p_i \triangleq \sum_{j=0}^{k_i} L_{i,j}^a L_{i,j+1}$, and $\text{vec}(V_{i,j})$ denotes the vectorization of $V_{i,j}$ performed in column-major order (i.e., the columns are stacked sequentially to form a vector). For all $(i,j) \in \{0, \ldots, b\} \times \{0, \ldots, k_i\}$, the ensemble activation function of the j^{th} layer of the i^{th} block is given by $\phi_{i,j} : \mathbb{R}^{L_{i,j}} \to \mathbb{R}^{L_{i,j}^a}$, where $\phi_{i,j}(\varphi_{i,j-1}) \triangleq \begin{bmatrix} \varsigma_{i,j}((\varphi_{i,j-1})_1) & \varsigma_{i,j}((\varphi_{i,j-1})_2) & \cdots & \varsigma_{i,j}\left((\varphi_{i,j-1})_{L_{i,j}}\right) & 1 \end{bmatrix}^{\top} \in \mathbb{R}^{L_{i,j}^a}$, the activation function is $\varsigma_{i,j} : \mathbb{R} \to \mathbb{R}$, and 1 accounts for the bias term.

A pre-activation design is used so that, before each block (except block 0), the output of the previous block is processed by an external activation function. Specifically, for each block $i \in \{1, \ldots, b\}$, define the pre-activation mapping $\psi_i : \mathbb{R}^{L_{i,k_i+1}} \to \mathbb{R}^{L_{i,k_i+1}^a}$ by $\psi_i(\kappa_i) = \begin{bmatrix} \varrho_i((\kappa_i)_1) & \varrho_i((\kappa_i)_2) & \cdots & \varrho_i\left((\kappa_i)_{L_{i,k_i+1}}\right) & 1 \end{bmatrix}^{\top} \in \mathbb{R}^{L_{i,k_i+1}^a}$ where $(\kappa_i)_\ell$ denotes the ℓ^{th} component of κ_i, the activation function is $\varrho_i : \mathbb{R} \to \mathbb{R}$, and 1 accounts for the bias term. The output of ψ_i serves as the input to block i and the residual connection is implemented by adding the current block output to the pre-activated output from the previous block. Hence, the ResNet recursion is defined by

$$\kappa_{i+1} \triangleq \begin{cases} \Phi_0(\kappa_0^a, \theta_0), & i = 0, \\ \kappa_i + \Phi_i(\psi_i(\kappa_i), \theta_i), & i \in \{1, \ldots, b\}, \end{cases} \tag{2}$$

with output $y \in \mathbb{R}^{L_{\text{out}}}$ and overall parameter vector $\Theta \triangleq \begin{bmatrix} \theta_0^{\top} & \cdots & \theta_b^{\top} \end{bmatrix}^{\top} \in \mathbb{R}^{\mathtt{p}}$, with $\mathtt{p} \triangleq \sum_{i=0}^{b} p_i$, where $\kappa_0^a \triangleq \begin{bmatrix} \kappa_0^{\top} & 1 \end{bmatrix}^{\top} \in \mathbb{R}^{L_{0,0}^a}$ denotes the augmented input to block $i = 0$. Therefore, the complete ResNet is represented as $\Psi : \mathbb{R}^{L_{\text{in}}} \times \mathbb{R}^{\mathtt{p}} \to \mathbb{R}^{L_{\text{out}}}$ expressed as $\Psi(x, \Theta) = \kappa_{b+1}$.

In order for ResNets to act as are universal function approximators of continuous functions on compact domains, their activation functions must be selected to belong to a class of functions called $\mathcal{A}_{\text{quad}}$.

Definition 1. *[?, Definition 3.1] The activation function $\varsigma : \mathbb{R} \to \mathbb{R}$ is said to be of class $\mathcal{A}_{quad}$ (denoted $\varsigma \in \mathcal{A}_{quad}$) if it is (i)*

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nondecreasing, (ii) non-constant, and (iii) there exists an $\ell \in \mathbb{Z}_{\geq 0}$ such that ς is a globally-defined solution to $\frac{d\zeta}{ds} = a_0 + a_1\zeta + a_2\zeta^2$, where $\zeta \triangleq \frac{d^{(\ell)}\varsigma}{ds^{(\ell)}}$ is injective, $a_0, a_1, a_2 \in \mathbb{R}$, $a_2 \neq 0$, and $\frac{d^{(0)}\varsigma}{ds^{(0)}} = \varsigma$.[1]

Theorem 1. *[?, Corollary 5.2] (Universal Approximation Theorem for Residual Neural Networks). For any compact domain $\Omega \subsetneq \mathbb{R}^{L_{in}}$, continuous function $F : \Omega \to \mathbb{R}^{L_{out}}$, and approximation accuracy $\overline{\varepsilon} \in \mathbb{R}_{>0}$, there exists a number of blocks $b \in \mathbb{Z}_{>0}$ (thus, a $\mathtt{p} \in \mathbb{Z}_{>0}$) and a parameter vector $\theta^* \in \mathbb{R}^{\mathtt{p}}$ such that for a ResNet architecture $\Psi : \mathbb{R}^{L_{in}} \times \mathbb{R}^{\mathtt{p}} \to \mathbb{R}^{L_{out}}$ with blocks of width $w = \max(L_{in}, L_{out}) + 1$ (i.e., $L_{i,j} \geq w$ for all $(i,j) \in \{0,\ldots,b\} \times \{1,\ldots,k_i\}$) and depth 1 (i.e., $k_i \geq 1$ for all $i \in \{0,\ldots,b\}$),*

$$\sup_{x\in\Omega} \|\varepsilon(x)\| \leq \overline{\varepsilon}, \tag{3}$$

where $\varepsilon(x) \triangleq h(x) - \Psi(x, \theta^*)$, provided that for all $(i,j) \in \{0,\ldots,b\} \times \{0,\ldots,k_i\}$, the activation functions $\varsigma_{i,j} \in \mathcal{A}_{quad}$.

HAVE TO MENTION HIGHER-ORDER APPROXIMATION.

III. Problem Formulation

Consider the second-order nonlinear dynamical system described by the differential equation,

$$\ddot{q}(t) = f(q(t), \dot{q}(t)) + u(t) + d(t), \tag{4}$$

where $t \in \mathbb{R}_{\geq 0}$ denotes time, $u : \mathbb{R}_{\geq 0} \to \mathbb{R}^n$ denotes the control input, $f : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^n$ denotes unknown drift dynamics of class $\mathcal{C}^2$, and $d : \mathbb{R}_{\geq 0} \to \mathbb{R}^n$ denotes an unknown, time-variant exogenous disturbance of class $\mathcal{C}^2$. For an initial condition set $(t_0, q_0, \dot{q}_0) \in \mathbb{R}_{\geq 0} \times \mathbb{R}^n \times \mathbb{R}^n$, let $q : \mathcal{I} \to \mathbb{R}^n$, $\dot{q} : \mathcal{I} \to \mathbb{R}^n$, and $\ddot{q} : \mathcal{I} \to \mathbb{R}^n$ denote the generalized position, velocity, and acceleration, respectively, of the resulting solution to (4) such that $q(t_0) = q_0$ and $\dot{q}(t_0) = \dot{q}_0$ and whose interval of existence is $\mathcal{I} \triangleq [0, T_{\max})$ for $T_{\max} \in \mathbb{R}_{>t_0} \cup \{\infty\}$.

The desired trajectory for (4) is governed by the autonomous second-order system,

$$\ddot{q}_d(t) = h(q_d(t), \dot{q}_d(t)), \tag{5}$$

where $h : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^n$ is an unknown function of class $\mathcal{C}^2$. For an initial condition set $(t_0, q_{d,0}, \dot{q}_{d,0}) \in \mathbb{R}_{\geq 0} \times \mathbb{R}^n \times \mathbb{R}^n$ and $q_d : \mathbb{R}_{\geq t_0} \to \mathbb{R}^n$, $\dot{q}_d : \mathbb{R}_{\geq t_0} \to \mathbb{R}^n$ and $\ddot{q}_d : \mathbb{R}_{\geq t_0} \to \mathbb{R}^n$ denote generalized desired position, velocity, and acceleration, respectively.

The control objective is to design a controller such that q converges exponentially to q_d despite the presence of non-vanishing disturbances, unknown dynamics, and lack of knowledge of $\ddot{q}_d$. The subsequent assumptions ensure feasibility of this objective.

Assumption 1. [2, Assumption 1] There exist known constants $\overline{q}_d, \overline{\dot{q}}_d, \overline{\ddot{q}}_d, \overline{\dddot{q}}_d \in \mathbb{R}_{\geq 0}$ such that $\|q_d(t)\| \leq \overline{q}_d$, $\|\dot{q}_d(t)\| \leq \overline{\dot{q}}_d$, $\|\ddot{q}_d(t)\| \leq \overline{\ddot{q}}_d$, and $\|\dddot{q}_d(t)\| \leq \overline{\dddot{q}}_d$ for all $t \in \mathbb{R}_{\geq 0}$.

Assumption 2. [2] There exist known constants $\overline{d}, \overline{\dot{d}}, \overline{\ddot{d}} \in \mathbb{R}_{\geq 0}$ such that $\|d(t)\| \leq \overline{d}$, $\left\|\dot{d}(t)\right\| \leq \overline{\dot{d}}$, and $\left\|\ddot{d}(t)\right\| \leq \overline{\ddot{d}}$ for all $t \in \mathbb{R}_{\geq 0}$.[2]

IV. Control Design

For subsequent analysis, let the tracking component of the controller obtained from feedback linearization $u_d : \mathbb{R}_{\geq 0} \to \mathbb{R}^n$ as

$$u_d(t) \triangleq d(t) + f(q_d(t), \dot{q}_d(t)) - \ddot{q}_d(t). \tag{6}$$

Since f is $\mathcal{C}^1$, Assumption 1 and extreme value theorem (EVT) guarantee that there exist constants $\overline{f}, \overline{\dot{f}} \in \mathbb{R}_{\geq 0}$ such that $\|f(q_d(t), \dot{q}_d(t))\| \leq \overline{f}$ and $\left\|\dot{f}(q_d(t), \dot{q}_d(t))\right\| \leq \overline{\dot{f}}$ for all $t \in \mathbb{R}_{\geq 0}$.

Assumption 3. The constants $\overline{f}, \overline{\dot{f}} \in \mathbb{R}_{\geq 0}$ are known.

By Assumption 3, there exist known bounds $\overline{u}_d \triangleq \overline{d} + \overline{f} + \overline{\ddot{q}}_d$ and $\overline{\dot{u}}_d \triangleq \overline{\dddot{q}}_d + \overline{\dot{f}}\overline{\dot{q}}_d + \overline{\dot{f}}\overline{\ddot{q}}_d + \overline{\dot{d}}$ such that $\|u_d(t)\| \leq \overline{u}_d$ and $\|\dot{u}_d(t)\| \leq \overline{\dot{u}}_d$ for all $t \in \mathbb{R}_{\geq 0}$. Subsequent analysis requires a bound on the difference $\widetilde{u} \triangleq u_d - u$ in terms of the ensemble system state $z : \mathcal{I} \to \mathbb{R}^{3n}$ defined as $z(t) \triangleq \begin{bmatrix} e^\top(t) & r_1^\top(t) & r_2^\top(t) \end{bmatrix}^\top$. Since f is $\mathcal{C}^2$, EVT, Assumption 1, and [4, Lemma 4] guarantee that there exist functions $\rho_f, \rho_{f'} : \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ that are class $\mathcal{P}_\infty$ with respect to each argument such that $\|f(q,\dot{q}) - f(q_d, \dot{q}_d)\| \leq \rho_f(k_1, \|z\|)\|z\|$ and $\left\| \left.\frac{\partial f}{\partial \dot{q}}\right|_{(q_d,\dot{q}_d)} - \left.\frac{\partial f}{\partial \dot{q}}\right|_{(q,\dot{q})} \right\| \leq \rho_{f'}(k_1, \|z\|)\|z\|$.

Assumption 4. The functions ρ_f and $\rho_{f'}$ are known.

Subtracting (6) from (4) and using (8), (9), and the time derivative of (9) gives that

$$\begin{aligned} \widetilde{u}(t) &= r_2(t) - k_1\dot{e}(t) - k_2 r_1(t) \\ &\quad - f(q_d(t), \dot{q}_d(t)) + f(q(t), \dot{q}(t)), \end{aligned} \tag{7}$$

for all $t \in \mathcal{I}$. By Assumption 4, there exists a known function $\rho_{\widetilde{u}} : \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$that is of class $\mathcal{P}_\infty$ with respect to each argument such that $\|\widetilde{u}\| \leq \rho_{\widetilde{u}}(k_1, k_2, \|z\|)\|z\|$ where $\rho_{\widetilde{u}}(k_1, k_2, s) \triangleq 1 + k_1 + k_1^2 + k_2 + \rho_f(k_1, s)$.

To facilitate subsequent control design, the first and second auxiliary tracking errors are defined as

$$r_1(t) \triangleq \dot{e}(t) + k_1 e(t), \tag{8}$$

$$r_2(t) \triangleq \dot{r}_1(t) + k_2 r_1(t), \tag{9}$$

where $k_1, k_2 \in \mathbb{R}_{>0}$ are user-selected gains. Differentiating (8) and using (4), (5), and (8) gives

$$\begin{aligned} \dot{r}_1(t) &= h(q_d(t), \dot{q}_d(t)) - f(q(t), \dot{q}(t)) \\ &\quad - u(t) - d(t) + k_1 r_1(t) - k_1^2 e(t), \end{aligned} \tag{10}$$

for all $t \in \mathcal{I}$. Differentiating (9) and substituting (10) and (8) gives

$$\begin{aligned} \dot{r}_2(t) &= F(\kappa(t)) - \left.\frac{\partial f}{\partial \dot{q}}\right|_{(q(t),\dot{q}(t))} d(t) - \dot{u}(t) - \dot{d}(t) \\ &\quad + (k_1 + k_2) r_2(t) - \left(k_1 k_2 + k_2^2 + k_1^2\right) r_1(t) + k_1^3 e(t), \end{aligned} \tag{11}$$

[1]By consequence of Definition 1, any $\varsigma \in \mathcal{A}_{\text{quad}}$ is necessarily an analytical function. Thus, ς and all its derivatives are locally Lipschitz on $\mathbb{R}$. Hyperbolic tangent, sigmoid, and softplus are of class $\mathcal{A}_{\text{quad}}$ but ReLU, swish, and the identity mapping are not. The function $\varsigma(s) = \frac{1}{c}\log(1 + \mathrm{e}^{cs})$ approximates ReLU for large $c \in \mathbb{R}_{>0}$ and is of class $\mathcal{A}_{\text{quad}}$. Leaky ReLU is given as $\varsigma(s) = s$ for all $s \in \mathbb{R}_{\geq 0}$ and $\varsigma(s) = rs$ (with $r \in \mathbb{R}_{>0}$) for all $s < 0$. The function $\varsigma(s) = rs + \frac{1}{c}\log\left(1 + \mathrm{e}^{(1-r)cs}\right)$ approximates leaky ReLU for large $c \in \mathbb{R}_{>0}$ and is of class $\mathcal{A}_{\text{quad}}$.

[2]While adaptive RISE techniques like [3] could be used to compensate for unknown bounds $\overline{d}, \overline{\dot{d}}, \overline{\ddot{d}}$, the described formulation therein yields asymptotic convergence. Achieving exponential convergence (as desired in this work) without Assumption 2 remains an open challenge.

for all $t \in \mathcal{I}$, where $\kappa : \mathcal{I} \to \mathbb{R}^{5n}$ is defined as $\kappa \triangleq \begin{bmatrix} q^\top & \dot{q}^\top & q_d^\top & \dot{q}_d^\top & u^\top \end{bmatrix}^\top$ and $F : \mathbb{R}^{5n} \to \mathbb{R}^n$ is defined as $F(\kappa) \triangleq \left.\frac{\partial h}{\partial q_d}\right|_{(q_d,\dot{q}_d)} \dot{q}_d + \left.\frac{\partial h}{\partial \dot{q}_d}\right|_{(q_d,\dot{q}_d)} h(q_d,\dot{q}_d) - \left.\frac{\partial f}{\partial q}\right|_{(q,\dot{q})} \dot{q} - \left.\frac{\partial f}{\partial \dot{q}}\right|_{(q,\dot{q})} f(q,\dot{q}) - \left.\frac{\partial f}{\partial \dot{q}}\right|_{(q,\dot{q})} u$.

A. Function Approximation

Since f and h are unknown, F is unknown and will be approximated online with a user-selected parametric universal approximator as $\Phi\left(\kappa, \widehat{\theta}\right)$ where $\Phi : \mathbb{R}^{L_{\text{in}}} \times \mathbb{R}^{\mathtt{p}} \to \mathbb{R}^{L_{\text{out}}}$ is a function approximator architecture mapping of class $\mathcal{C}^2$, $\widehat{\theta} \in \mathbb{R}^{\mathtt{p}}$ is a parameter, $L_{\text{out}} = n$, $L_{\text{in}} = 5n$, and $\mathtt{p} \in \mathbb{Z}_{\geq 1}$ is the number of parameters.

Remark 1. In this work, any class $\mathcal{C}^2$ parametric approximator which satisfies a universal approximation theorem like Theorem 1 can be used for Φ, in which there must exist a sufficiently sized architecture (e.g., network width for shallow neural networks, network depth for deep neural networks of fixed width, or number of blocks for ResNets) and a corresponding ideal parameter $\theta^* \in \mathbb{R}^{\mathtt{p}}$ such that any function of class $\mathcal{C}^1$ can be approximated to arbitrarily small precision under the C^1 norm on a compact set $\Omega \subsetneq \mathbb{R}^{L_{\text{in}}}$. For example, radial basis functions, Chebyshev polynomials, transformers, or Kolmogorov-Arnold networks can be used.

To construct the compact domain of inputs over which the approximation of F by $\Phi\left(\kappa, \widehat{\theta}\right)$ is of interest, the idealized input $\kappa_d : \mathbb{R}_{\geq 0} \to \mathbb{R}^{5n}$ seen when (4) perfectly tracks (5) is defined as $\kappa_d \triangleq \begin{bmatrix} q_d^\top & \dot{q}_d^\top & q_d^\top & \dot{q}_d^\top & u_d^\top \end{bmatrix}^\top \in \mathbb{R}^{5n}$. By Assumptions 1 and 3, there exist known constants $\overline{\kappa}_d \triangleq 2\overline{q}_d + 2\overline{\dot{q}}_d + \overline{u}_d$ and $\overline{\dot{\kappa}}_d \triangleq 2\overline{\dot{q}}_d + 2\overline{\ddot{q}}_d + \overline{\dot{u}}_d$ such that $\|\kappa_d\| \leq \overline{\kappa}_d$ and $\|\dot{\kappa}_d\| \leq \overline{\dot{\kappa}}_d$ for all $t \in \mathbb{R}_{\geq 0}$. Using (8), Assumption 4, and the definition of $\rho_{\widetilde{u}}$, there exists a known function that is of class $\mathcal{P}_\infty$ with respect to each argument such that $\|\widetilde{\kappa}\| \leq \rho_{\widetilde{\kappa}}(k_1, k_2, \|z\|)\|z\|$, where $\widetilde{\kappa} \triangleq \kappa_d - \kappa$ and $\rho_{\widetilde{\kappa}}(k_1, k_2, s) \triangleq k_1 + 2 + \rho_{\widetilde{u}}(k_1, k_2, s)$. The approximation domain of interest for all inputs κ is given as

$$\Omega \triangleq \left\{\zeta \in \mathbb{R}^{5n} : \|\zeta\| \leq \overline{\kappa}\right\}, \quad \overline{\kappa} \triangleq \overline{\kappa}_d + \rho_{\widetilde{\kappa}}(k_1, k_2, C_\Delta) C_\Delta, \tag{12}$$

where $C_\Delta \in \mathbb{R}_{>0}$ is a subsequently defined constant.

The parameter $\widehat{\theta}$ is intended to dynamically estimate θ^* and evolve continuously but be constrained to a user-defined bounded search space $\mho$. Without loss of generality, it is assumed herein that $\mho = \overline{B}_{\overline{\theta}}(0_{\mathtt{p}})$, where $\overline{\theta} \in \mathbb{R}_{>0}$ is a user-defined radius. The rule governing the trajectory of $\widehat{\theta}$ over time is called the update law and is chosen by the user. The update law must satisfy the following properties.

Property 1. [5, Assumption 1] The parameter's trajectory $\widehat{\theta} : \mathbb{R}_{\geq 0} \to \mathbb{R}^{\mathtt{p}}$ is of class $\mathcal{C}^1$ and there exists known constants $\overline{\theta}, \eta_1 \in \mathbb{R}_{>0}$ such that the trajectory and its first time derivative $\dot{\widehat{\theta}} : \mathbb{R}_{\geq 0} \to \mathbb{R}^{\mathtt{p}}$ satisfy $\left\|\widehat{\theta}(t)\right\| \leq \overline{\theta}$ and $\left\|\dot{\widehat{\theta}}(t)\right\| \leq \eta_1$ for all $t \in \mathbb{R}_{\geq 0}$.

Property 2. There exists a known constant $\overline{\varepsilon}_{\mathrm{d}} \in \mathbb{R}_{>0}$ such that parameter's trajectory satisfies $E_\Omega\left(\widehat{\theta}(t)\right) \leq \overline{\varepsilon}_{\mathrm{d}}$ for all $t \in \mathbb{R}_{\geq 0}$, where $\varepsilon(\kappa, \theta) \triangleq \Phi(\kappa, \theta) - F(\kappa)$ and $E_\Omega(\theta) \triangleq \sup_{\kappa \in \Omega} \|\varepsilon(\kappa, \theta)\| + \sup_{\kappa \in \Omega} \left\| \left.\frac{d\varepsilon}{d\kappa}\right|_{(\kappa,\theta)} \right\|$.

Property 1 can be met, for instance, by using a projection algorithm and limiting the speed of parameter changes. Satisfying Property 2 requires careful consideration of both parameter initialization and the update law. The initial parameter $\widehat{\theta}(t_0)$ must either be obtained by pre-training the function approximator on input-output pairs generated by a function representative of F or by using the following approach. Notice that EVT ensures that $\|F\| + \left\|\frac{dF}{d\kappa}\right\|$ achieves a finite maximum over Ω since Ω is compact and F is of class $\mathcal{C}^1$. Assume a bound $\overline{F} \in \mathbb{R}_{\geq 0}$ on such a maximum is known. Pick $\widehat{\theta}(t_0)$ and a suitable $\mho$ such that $\Phi\left(\kappa, \widehat{\theta}(t_0)\right) = 0_n$ for all $\kappa \in \Omega$. Finally, select $\overline{\varepsilon}_{\mathrm{d}} = \overline{F}$. Satisfying Property 2 beyond the initial time requires that any information gained online to refine the parameter not result in a $E_\Omega\left(\widehat{\theta}(t)\right)$ that exceeds $\overline{\varepsilon}_{\mathrm{d}}$. Heuristically, this phenomena, called catastrophic forgetting, can be avoided by using experience replay methods like concurrent learning (CL).[3]

Since the expressive powers of ResNets and many similar approximators rely on the ability to select θ^* from the whole of $\mathbb{R}^{\mathtt{p}}$ (cf. Theorem 1) rather than a prescribed $\mho$, F must belong to a strict subset of all continuous functions as a consequence of Property 1.

Assumption 5. The function F and its first derivative $\frac{dF}{d\kappa}$ can be uniformly approximated to arbitrary precision using bounded weights for a sufficiently large approximator architecture. Formally, $F \in \{\mathcal{C}^1(\Omega; \mathbb{R}^n) : \forall \overline{\varepsilon} \in \mathbb{R}_{>0}, \exists \mathtt{p} \in \mathbb{Z}_{\geq 1} : \exists \mho \subsetneq \mathbb{R}^{\mathtt{p}} : \exists \theta^* \in \mho : E_\Omega(\theta^*) \leq \overline{\varepsilon}\}$.[4]

Assumption 5 is required because of the following tradeoff. In parametric nonlinear regression, the user is forced to pick two of the following three desirable qualities for their approximator: (i) F can be any $\mathcal{C}^1$ function, (ii) F can be approximated arbitrarily well ($\overline{\varepsilon}_{\mathrm{d}} \to 0$ as $\mathtt{p} \to \infty$), (iii) a known bound exists on the weights ($\overline{\theta}$) needed to achieve the desired accuracy for the target function. Without (ii), there will exist sufficiently small values of $\overline{\varepsilon}_{\mathrm{d}}$ that are not achievable for the given $\mho$ even as $\mathtt{p} \to \infty$. That is, the infimum of all values $\overline{\varepsilon}_{\mathrm{d}}$ may achieve is an unknown positive value (rather than zero) that might be arbitrarily large. Thus, (ii) is required. Since computers possess finite memory, (iii) is required. Thus, (i) must be sacrificed in place of Assumption 5.

By Assumption 5, the set $\Theta^* \triangleq \arg\min_{\vartheta \in \mho} E_\Omega(\theta)$ is nonempty and all $\theta^* \in \Theta^*$ satisfy $\|\theta^*\| \leq \overline{\theta}$. Since the design of the update law and the choice of parametric approximator for Φ are at the user's discretion, the approach in this work is said to be modular.

B. Control Law

Motivated by the form of (11), the controller is designed as

$$\begin{aligned}\dot{u}(t) &= (k_1 + k_2 + k_3) r_2(t) + \left(1 - k_1^2 - k_2^2 - k_1 k_2\right) r_1(t) \\ &\quad + k_1^3 e(t) + K_{\text{RISE}} \operatorname{sgn}(r_1(t)) + \Phi\left(\kappa(t), \hat{\theta}(t)\right),\end{aligned} \tag{13}$$

[3]For example, a first-order CL-based update law for parameters in universal approximators like Φ can be found in [6]. Note that any discrete updates to a history stack made online may not satisfy the $\mathcal{C}^1$ requirement of Property 1. However, online history stack management is not required to prevent catastrophic forgetting.

[4]Assumption 5 is simply a formal statement of an implicit assumption common to many previous adaptive controls works using parametric approximators.

where $K_{\text{RISE}}, k_3 \in \mathbb{R}_{>0}$ are user-defined constants. Integrating both sides of (13) and picking the initial condition $u(t_0) = K_P e(t_0) + K_D \dot{e}(t_0)$ yields the implementable form

$$\begin{aligned} u(t) = K_P e(t) + K_D \dot{e}(t) + \int_{t_0}^{t} (K_I e(\tau))\, d\tau \\ + \left(\int_{t_0}^{t} K_{\text{RISE}} \operatorname{sgn}(r_1(\tau)) + \Phi\left(\kappa(\tau), \hat{\theta}(\tau)\right) \right) d\tau \end{aligned} \tag{14}$$

where $K_P \triangleq k_1k_2 + k_1k_3 + k_2k_3 + 1$, $K_I \triangleq k_1k_2k_3 + k_1$, and $K_D \triangleq k_1 + k_2 + k_3$.[5] Substituting (13) and the definition of ε into (11) and performing algebraic manipulation gives that $r_2 : \mathcal{I} \to \mathbb{R}^n$ is a Filippov solution to

$$\dot{r}_2(t) \in \widetilde{N}\left(\kappa, \kappa_d, \theta^*, \hat{\theta}, t\right) - k_3 r_2(t) - r_1(t) - K_{\text{RISE}} K[\operatorname{sgn}](r_1(t)) + D\left(\kappa_d, \theta^*, \hat{\theta}, t\right)$$

where

$$\begin{aligned} D\left(\kappa_d, \theta^*, \hat{\theta}, t\right) &\triangleq -\dot{d}(t) - \left.\frac{\partial f}{\partial \dot{q}}\right|_{(q_d, \dot{q}_d)} d(t) + \varepsilon(\kappa_d) + N_0\left(\kappa_d, \theta^*, \hat{\theta}\right), \\ N_0\left(\kappa, \theta^*, \hat{\theta}\right) &\triangleq \Phi(\kappa, \theta^*) - \Phi\left(\kappa, \hat{\theta}\right), \\ \widetilde{N}\left(\kappa, \kappa_d, \theta^*, \hat{\theta}, t\right) &\triangleq \widetilde{N}_1\left(\kappa, \kappa_d, \theta^*, \hat{\theta}\right) + \widetilde{N}_\varepsilon(\kappa, \kappa_d) + \widetilde{N}_2(q, \dot{q}, q_d, \dot{q}_d)\, d(t), \\ \widetilde{N}_\varepsilon(\kappa, \kappa_d) &\triangleq \varepsilon(\kappa) - \varepsilon(\kappa_d), \\ \widetilde{N}_1\left(\kappa, \kappa_d, \theta^*, \hat{\theta}\right) &\triangleq N_0\left(\kappa, \theta^*, \hat{\theta}\right) - N_0\left(\kappa_d, \theta^*, \hat{\theta}\right), \\ \widetilde{N}_2(q, \dot{q}, q_d, \dot{q}_d) &\triangleq \left.\frac{\partial f}{\partial \dot{q}}\right|_{(q_d, \dot{q}_d)} - \left.\frac{\partial f}{\partial \dot{q}}\right|_{(q, \dot{q})} \end{aligned}$$

Since Φ is of class $\mathcal{C}^1$ and κ_d is bounded, using [4, Lemma 4], the bound on $\|\widetilde{\kappa}\|$ and Property 5, there exists a known function $\rho_\Phi : \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ that is of class $\mathcal{P}_\infty$ with respect to each argument such that $\left\|\widetilde{N}_1\left(\kappa, \kappa_d, \theta^*, \hat{\theta}\right)\right\| \leq \rho_\Phi(k_1, k_2, \|z\|)\|z\|$. Since $\overline{\kappa}_d \leq \overline{\kappa}$, Property 2 ensures that $\|\varepsilon(\kappa_d(t))\| \leq \overline{\varepsilon}_{\mathrm{d}}$ and $\left\| \left.\frac{d\varepsilon}{d\kappa}\right|_{\kappa_d(t)} \right\| \leq \overline{\varepsilon}_{\mathrm{d}}$ for all $t \in \mathbb{R}_{\geq 0}$. Since $\|\kappa\| \leq \overline{\kappa}_d + \|\widetilde{\kappa}\|$, the bound on $\|\widetilde{\kappa}\|$ gives that $\|\kappa\| \leq \overline{\kappa}$ for all $\|z\| \leq C_\Delta$. Since ε is of class $\mathcal{C}^1$, it follows that $\left\|\widetilde{N}_\varepsilon(\kappa, \kappa_d)\right\| \leq \overline{\varepsilon}_{\mathrm{d}} \rho_{\widetilde{\kappa}}(k_1, k_2, \|z\|)\|z\|$ for all $\|z\| \leq C_\Delta$ [4, Lemma 4]. Since f is of class $\mathcal{C}^2$, Assumption 1 and EVT guarantee that there exist constants $\overline{f'}, \overline{f''} \in \mathbb{R}_{\geq 0}$ such $\left\| \left.\frac{\partial f}{\partial \dot{q}}\right|_{(q_d(t), \dot{q}_d(t))} \right\| \leq \overline{f'}$, $\left\| \left.\frac{\partial f}{\partial \dot{q} \partial q}\right|_{(q_d, \dot{q}_d)} \right\| \leq \overline{f''}$, and $\left\| \left.\frac{\partial^2 f}{\partial \dot{q}^2}\right|_{(q_d, \dot{q}_d)} \right\| \leq \overline{f''}$ for all $t \in \mathbb{R}_{\geq 0}$.

Assumption 6. The constants $\overline{f'}$ and $\overline{f''}$ are known.

Since Φ is of class $\mathcal{C}^0$ and $\|\kappa_d(t)\|$, $\|\dot{\kappa}_d(t)\|$, $\|\varepsilon(\kappa_d(t))\|$, and $\left\| \left.\frac{d\varepsilon}{d\kappa}\right|_{\kappa_d(t)} \right\|$ are bounded by known constants, EVT, Property 5, and Assumptions 2 and 6 ensure that there exist known constants $\overline{D}, \overline{\dot{D}} \in \mathbb{R}_{\geq 0}$ such that $\left\|D\left(\kappa_d, \theta^*, \hat{\theta}, t\right)\right\| \leq \overline{D}$ and $\left\|\dot{D}\left(\kappa_d, \theta^*, \hat{\theta}, t\right)\right\| \leq \overline{\dot{D}}$ for all $t \in \mathbb{R}_{\geq 0}$. By Assumption 4, $\left\|\widetilde{N}_2(q, \dot{q}, q_d, \dot{q}_d)\right\| \leq \rho_{f'}(k_1, \|z\|)\|z\|$. Using Assumption 2, it follows that there exists a known function $\rho_0 : \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ that is of class $\mathcal{P}_\infty$ with respect to each argument such that $\left\|\widetilde{N}\left(\kappa, \kappa_d, \theta^*, \hat{\theta}, t\right)\right\| \leq \rho_0(k_1, k_2, \|z\|)\|z\|$ for all $\|z\| \leq C_\Delta$, where $\rho_0(k_1, k_2, s) \triangleq \rho_\Phi(k_1, k_2, s) + \overline{\varepsilon}_{\mathrm{d}} \rho_{\widetilde{\kappa}}(k_1, k_2, s) + \overline{d} \rho_{f'}(k_1, s)$.

[5] Although $\ddot{q}$ is not assumed to be measurable, it appears in (13) by virtue of r_2. However, note that the implementable form only requires q and $\dot{q}$.

V. Stability Analysis

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